

What the Prices Are Hiding

A structural analysis of nine instruments over one simulated year — testing each for mean reversion, momentum, volatility clustering, lead-lag relationships and cross-sectional reversal, and converting what survives into trading models.

Data Round 1 · 365 days · 9 instruments

Methods ADF · Hurst · ACF & Ljung–Box · variance ratio · ARCH-LM & GARCH(1,1) · Granger · Engle–Granger · periodogram

Validation Split-half stability on every reported signal

Written for Readers with no prior finance or statistics background

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How to read this report. Every section is built in four layers, and you can stop at whichever one you need.

IN PLAIN ENGLISH What the test is asking, using no notation. If you read only these, you will still follow the whole argument.

THE MATHS The formula, with every symbol defined and — where it isn't obvious — where the formula comes from.

WORKED EXAMPLE The same arithmetic done on our actual data, so you can check it by hand.

WHAT IT MEANS FOR TRADING The decision the result leads to.

Nothing here is illustrative. Every figure, table and worked example is generated from the Round 1 CSVs by the scripts listed in Appendix C.

Executive summary

We were given one simulated year — 365 days of prices for nine instruments — and asked to find whatever structure is hiding in it, so we can bet on the second year we've never seen.

The honest headline is that **most of the board is noise**. Five of the nine instruments behave close to a coin flip from one day to the next. But three of them are not random at all, and two of those are extremely predictable:

STRONGEST EDGE

Sausage Sizzle

Its price is a *recipe*: bread and sausage prices from yesterday, plus a slow-moving labour cost. Since today's bread and sausage prices are already printed, tomorrow's sizzle price is largely knowable today. Correlation between our forecast and what actually happens: **0.96**. We get the direction right **83%** of the time.

SECOND STRONGEST

UQ Dollar

Pinned to \$100 by something that acts like a spring. Whenever it drifts away it is yanked back — and it reverts almost fully in a single day. Correlation between today's gap and tomorrow's move: **0.69**. When the gap is large we are right **92-95%** of the time.

THIRD

MenuDash

A blurry, rounded readout of the same hidden labour cost that the sizzle reveals exactly. When the two disagree, MenuDash closes the gap within about three days. Correlation: **0.28**.

Beyond those, Boat Party Ticket, Bread, Sausage and Fintech Token each carry a small but real edge (correlations of 0.13 to 0.21). Thrifted Jeans and Liferaft Ticket carry none we could verify.

Three findings contradict what the current algorithm assumes, and each would cost money in Round 2:

1. **Fintech Token's signal is fitted to noise.** The algorithm averages 5-, 7- and 10-day reversion windows. All three work in one half of the year and fail in the other — they flip sign. Only a 1-2 day window survives.
2. **The hard-coded Boat Party seasonal calendar does not transfer.** Fitting the day-of-year shape on the first half of Round 1 and applying it to the second half gives a correlation of **-0.12** — it points the wrong way. Round 1 contains two semesters and they are not copies of each other.
3. **Liferaft Ticket has no history to learn from.** Its price moves take exactly two values, $\pm \$5,000$ and $+\$8,000$, because it is a voting payoff rather than a price. Round 1's path records what Round 1's room did; it says nothing about Round 2's room. The contrarian rule tests at a correlation of -0.04 , which is indistinguishable from zero.

We also tested for **cross-sectional mean reversion** — the idea that today's biggest winner tends to be tomorrow's loser. At first glance it looks strongly present. It isn't. The apparent effect is one instrument (UQ Dollar) leaking into the panel statistic. Remove it and the effect disappears. **Do not build a rank-and-reverse book.**

Part I • The problem, and the one number that measures success

1. What we are actually trying to do

Nine instruments trade for 365 days. Each day we choose how many units of each to hold — long (betting the price rises) or short (betting it falls) — subject to per-instrument limits and a \$600,000 total budget. We are then scored on a *second* year we never get to see.

So this is not a question of "what happened in Round 1". It is: **what mechanism generated Round 1, and will that mechanism still be running in Round 2?**

That distinction drives everything below. A pattern that describes Round 1 perfectly but has no mechanism behind it is worse than useless, because we will bet real budget on it.

The nine Round 1 price series

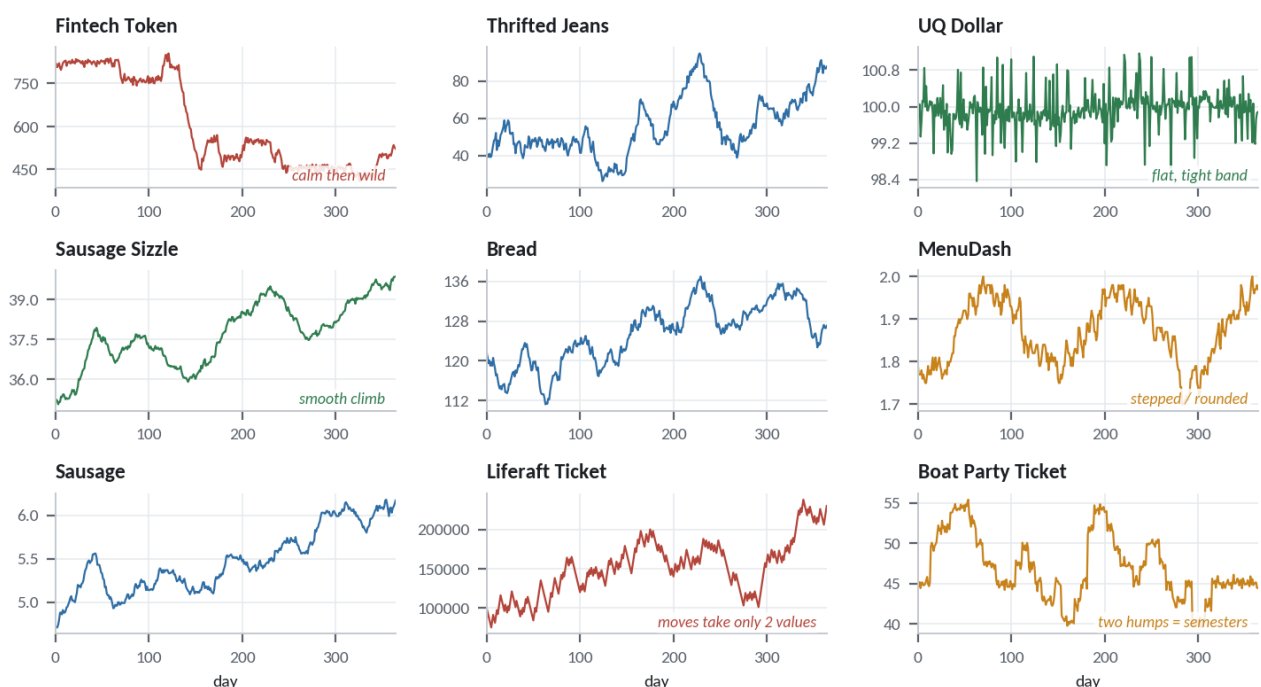


Figure 1. The nine Round 1 price series. Even by eye the board splits into groups: UQ Dollar is pinned flat inside a band of about $\pm \$1.60$; Sausage Sizzle climbs smoothly; MenuDash moves in visible steps; Boat Party has two clear humps; Fintech Token has calm stretches and violent ones. Liferaft drifts upward like the others here, but that is an artefact of the scale — its daily change only ever takes two values, which \$19 unpacks. Each of these visual impressions turns into a specific statistical test below.

2. The Information Coefficient — how we score every idea

IN PLAIN ENGLISH

Every trading idea, no matter how it's dressed up, boils down to producing one number each morning: a **signal**. Positive means "I think this goes up today", negative means "I think it goes down", and bigger means "I'm more confident".

There is one obvious way to grade such a thing: line up all 365 of your signals next to what the price actually did the following day, and measure how well they move together. That measure

is the correlation between the two, and in finance it has a name — the **Information Coefficient**, or IC.

An IC of 0 means your signal is worthless. An IC of 1 means you are psychic. Real tradeable signals in real markets typically live between 0.02 and 0.10; anything above 0.3 would be extraordinary. As we'll see, this competition's data is far more forgiving than a real market.

THE MATHS

For a signal s_t computed on day t using only information available on day t , and the price change $r_{t+1} = P_{t+1} - P_t$ that follows:

$$\text{IC} = \text{corr}(s_t, r_{t+1}) = \frac{\sum_t (s_t - \bar{s})(r_{t+1} - \bar{r})}{\sqrt{[\sum_t (s_t - \bar{s})^2 \cdot \sum_t (r_{t+1} - \bar{r})^2]}}$$

$\bar{s}, \bar{r}$ — the averages of the signal and of the next-day change.

The numerator asks: when the signal is above its average, is the next move also above *its* average? The denominator just rescales the answer onto $[-1, +1]$ so the units cancel and we can compare across instruments.

The critical detail is the subscript. The signal carries t , the return carries $t+1$. If any quantity dated $t+1$ or later sneaks into s_t , the IC becomes meaningless — you are measuring how well tomorrow predicts tomorrow. This is called *lookahead bias*, and §14 documents a case where we walked straight into it.

Turning an IC into a verdict. A correlation measured on a finite sample is itself uncertain. Under the null hypothesis that the true correlation is zero,

$$t = \text{IC} \cdot \frac{\sqrt{(n-2)}}{\sqrt{(1-\text{IC}^2)}}$$

follows a t -distribution with $n - 2$ degrees of freedom. The conventional threshold is $|t| > 2$, which for large n corresponds to roughly a 5% chance of seeing an effect this large from pure luck.

Rearranged, that gives the useful rule of thumb: with n observations, anything smaller than **1.96 / $\sqrt{n}$** is not distinguishable from noise.

WORKED EXAMPLE

We have $n = 365$ days, so our noise floor is

$$1.96 / \sqrt{364} = \mathbf{0.103}$$

Any IC below about 0.10 in this report should be treated as zero. Converting a few of our actual results:

Signal	IC	n	$t = IC \cdot \sqrt{(n-2)} / \sqrt{(1-IC^2)}$	Verdict
Sizzle: ingredients + drift	0.9601	353	64.3	overwhelming
UQ Dollar: reversion to peg	0.6975	348	18.1	overwhelming
MenuDash: fair-value gap	0.2801	284	4.90	strong
Fintech: 2-day reversion	0.1327	363	2.54	marginal
Thrifty Jeans: momentum	0.0971	350	1.82	not significant

Thrifty Jeans is the instructive one. An IC of 0.097 *looks* like a real pattern — but on 350 observations, a worthless signal produces something that large about 7% of the time by chance alone. That is not a standard we would bet budget on.

3. The split-half test — our defence against fooling ourselves

IN PLAIN ENGLISH

Here is the trap that ruins most backtests. We tried dozens of signal variants per instrument — reversion over 2 days, 3 days, ... 40 days, momentum over each of those, and so on. If you try forty things on random data, one or two will look brilliant purely by accident. Report the best one and you have discovered nothing except your own persistence.

The cheapest honest defence: **cut the year in half**. Measure the signal separately on days 0–181 and on days 182–364. A real mechanism should show up in both. A lucky accident usually shows up in one and vanishes — or reverses — in the other.

We required both halves to have the *same sign* and the full-sample t to exceed 2. This is a weaker check than a true out-of-sample test on data we've never touched — we can't do that, since Round 2 is hidden — but it catches the worst offenders, and it caught three.

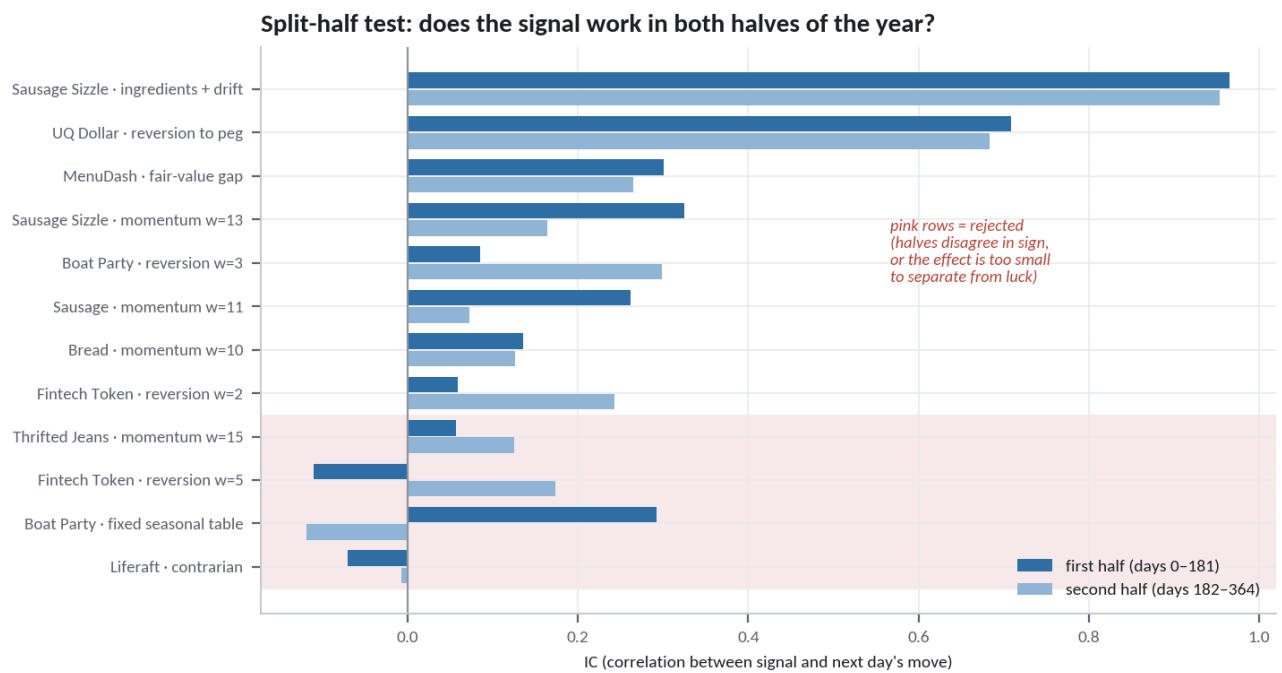


Figure 2. Every signal we considered, scored separately on each half of the year. The eight signals at the top hold up. The four in pink do not: Fintech's 5-day reversion and the Boat Party seasonal calendar actually *reverse sign* between halves, which is the signature of a pattern fitted to noise. Thrifty Jeans keeps its sign but is too weak to separate from luck.

WHAT IT MEANS FOR TRADING

Three signals currently in `algorithm.py` fail this test. Sections 12, 14 and 15 deal with each in turn. Every other recommendation in this report cleared it.

Part II • Does each series have structure at all?

Before asking *which* pattern an instrument follows, we should ask whether it follows any. Two tests answer that, and they approach it from opposite directions.

4. Stationarity — does the price have a home to return to?

IN PLAIN ENGLISH

Picture two objects. The first is a **balloon released outdoors**: it drifts, and where it goes next depends on where it is now, but it has no home. Knowing it is currently high tells you nothing about whether it will come down. The second is a **weight hanging on a spring**: push it anywhere you like and it gets pulled back to a fixed resting point, harder the further you push.

The balloon is a *random walk*. The spring is *mean-reverting*, or in the jargon *stationary*. This is the single most important thing to know about a price series, because it tells you which direction to bet after a large move — with it, or against it.

The test works by measuring the pull. Take each day's price change and ask: is it related to how far the price was from the average yesterday? For a balloon, no relationship. For a spring, a strongly negative one — high yesterday means a fall today.

THE MATHS

Fit the regression (this is the core of the **Augmented Dickey-Fuller test**):

$$\Delta P_t = a + b P_{t-1} + \varepsilon_t, \quad \text{where } \Delta P_t = P_t - P_{t-1}$$

b — the pull. This is the number that matters.

a — an intercept, which combined with b locates the resting point.

ε_t — the unpredictable shock on day t .

If $b = 0$ the change doesn't depend on the level at all: a random walk. If $b < 0$, being high causes a fall — a spring. The ADF test asks whether b is significantly below zero; a **p-value below 0.05 means mean-reverting**.

Rewriting the same equation makes its meaning clearer. Setting $\kappa = -b$ and $\mu = -a/b$:

$$\Delta P_t = \kappa(\mu - P_{t-1}) + \varepsilon_t$$

which reads directly as "each day, move a fraction κ of the way back toward the resting level μ ". This is the discrete-time **Ornstein-Uhlenbeck process**, the standard model for anything that reverts to a level.

Deriving the half-life. Ignore the shocks for a moment. If the gap from the resting point today is $g_t = P_t - \mu$, the equation says $g_{t+1} = (1 - \kappa)g_t$, so after h days the gap is $(1 - \kappa)^h g_0$. Ask when half the gap is gone:

$$(1 - \kappa)^h = 1/2 \implies h = \frac{\ln 1/2}{\ln(1 - \kappa)} \approx \frac{\ln 2}{\kappa} \quad (\text{for small } \kappa)$$

The **half-life** is the natural way to report reversion speed: it is how many days it takes to close half the gap, and it tells you directly how long you'd hold the trade.

WORKED EXAMPLE — UQ DOLLAR

Running that regression on all 364 daily changes:

$$\Delta P_t = 96.190 - 0.9625 P_{t-1} \quad (t \text{ on the slope} = -18.3)$$

So $\kappa = 0.9625$ and $\mu = 96.190 / 0.9625 = \mathbf{99.941}$.

Two things fall out. First, the resting point is \$99.94 — essentially the \$100 peg the instrument specification describes, recovered from the data alone. Second, $\kappa = 0.96$ means the price closes **96% of any gap in a single day**. The half-life is $\ln 2 / 0.9625 = \mathbf{0.72 \text{ days}}$.

That is extraordinarily fast. A typical mean-reverting pair in a real market has a half-life of weeks. This one is essentially fully reverted by tomorrow's close.

UQ Dollar: a spring pulling back to \$100

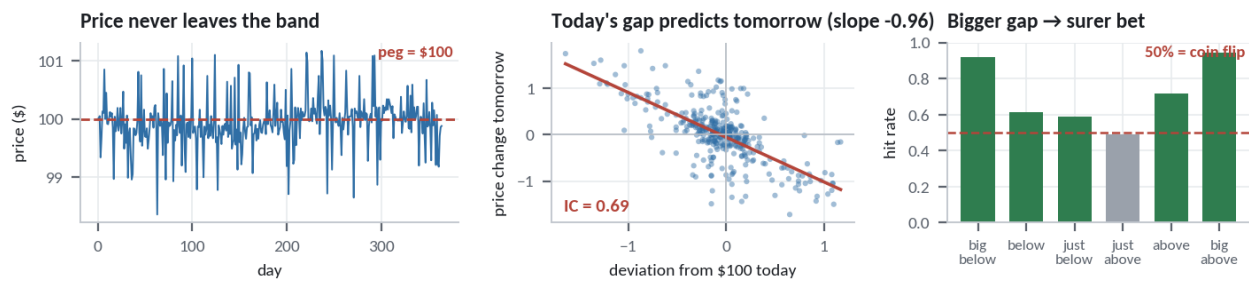


Figure 3. Left: the price never escapes a band of about $\pm \$1.60$ around \$100 in the whole year. Middle: each dot is one day — the horizontal axis is how far the price sat from \$100, the vertical axis is what it did the next day. The relationship is close to a straight line with slope -0.96 , which is exactly the spring constant we fitted. Right: the further from the peg, the more reliable the bet — from a coin flip when the gap is tiny to a 92–95% hit rate when it is large.

A note on the p-values. UQ Dollar's ADF p-value is 0.103 — technically above the 0.05 threshold, which would normally mean "cannot reject a random walk". Do not be misled. The ADF test on 365 observations has low power, and every other piece of evidence ($\kappa = 0.96$ with $t = -18.3$, $ACF(1) = -0.49$, $Hurst = 0.008$, variance ratio = 0.11) points the same way with overwhelming force. **When a single test disagrees with five others, the single test is usually the one with the problem.** We report the p-value for completeness and ignore it.

5. The Hurst exponent — wandering versus snapping back

IN PLAIN ENGLISH

Here is a second, cleverer way to ask the same question, and it needs no regression at all.

Take a series and measure how far it typically travels in 2 days. Then in 20 days. For a **balloon**, the distances grow like the square root of time — travel $10\times$ longer and you get about $\sqrt{10} \approx 3.2\times$ further. This is the same square-root law that governs a drunkard's walk, and it is what pure randomness looks like.

Now: if the series is a **spring**, it can't get far no matter how long you wait, because it keeps getting hauled back. Distance barely grows at all. And if the series **trends**, each step tends to continue the last, so distance grows *faster* than square-root.

Measuring how fast distance grows with time therefore diagnoses the series in one number.

THE MATHS

For each lag ℓ , compute the standard deviation of the ℓ -day change:

$$\tau(\ell) = \text{sd}(P_{t+\ell} - P_t)$$

The claim is that this scales as a power law, $\tau(\ell) \propto \ell^H$. Taking logs turns that into a straight line:

$$\ln \tau(\ell) = H \cdot \ln \ell + \text{constant}$$

so H — the **Hurst exponent** — is just the slope of a log-log plot, which we estimate by least squares over $\ell = 2 \dots 39$.

$H = 0.5$ — random walk. Distance grows as $\sqrt{\text{time}}$.

$H < 0.5$ — mean-reverting. The lower, the tighter the leash.

$H > 0.5$ — trending. Moves tend to continue.

WORKED EXAMPLE

Instrument	sd of 2-day change	sd of 20-day change	ratio	H
<i>a pure random walk would give</i>	—	—	3.16	0.50
UQ Dollar	0.602	0.638	1.06	0.008
Thrifty Jeans	3.818	15.219	3.99	0.576
Sausage Sizzle	0.143	0.758	5.32	0.719

UQ Dollar travels **no further in 20 days than it does in 2** — the ratio is 1.06 where randomness demands 3.16. That is about as strong a mean-reversion signature as this statistic can produce.

Sausage Sizzle goes the other way, travelling 5.3× rather than 3.2×. But be careful how you read that: its H of 0.72 reflects the slow upward march of ingredient costs, not a tradeable tendency for moves to continue. **A trending cause is not the same as a trending signal** — §11 shows the far better way to trade it.

Thrifty Jeans at 3.99 versus 3.16 is close enough to random to be treated as random.

6. Structural results for all nine

Half-life is from the OU fit; a value in the tens or hundreds of days on a one-year sample means "no meaningful reversion". Drift t tests whether the average daily change differs from zero.

Instrument	ADF p	Hurst	Half-life	Drift t	Ann. vol	What it is
UQ Dollar	0.10	0.008	0.72d	-0.01	9.5%	Hard peg — a spring at \$100
Sausage Sizzle	0.49	0.72	152d	2.52	4.1%	Deterministic cost function
MenuDash	0.16	0.52	22d	0.47	16.0%	Rounded readout of that cost
Boat Party	0.07	0.55	21d	-0.03	34.8%	Seasonal trend + overreaction
Fintech Token	0.56	0.58	129d	-1.11	37.8%	Random walk, volatility regimes
Bread	0.40	0.51	66d	0.32	11.2%	Random walk, weak trend
Sausage	0.84	0.56	134d	2.23	9.9%	Drifting random walk
Thrifty Jeans	0.15	0.58	59d	0.90	86.1%	Drifting random walk
Liferaft	—	0.45	—	1.06	73.8%	Not a price process (\$15)

Two entries deserve a second look.

Thrifty Jeans gained 119% over the year — the best performer on the board — yet its drift t is only 0.90. How can such a large gain not be significant? Because its annualised volatility is 86%. A series that swings that violently can produce a 119% year by luck without any genuine upward tendency. One year is simply not enough data to tell a real drift from a fortunate random walk. Betting on it is a bet, not a signal — and we label it as such.

Boat Party's ADF p-value of 0.07 is close to significant, but the reversion it picks up is the seasonal shape returning to its own average, not something we can trade at the daily horizon. \$14 finds what is tradeable there.

WHAT IT MEANS FOR TRADING

Only UQ Dollar earns a genuine mean-reversion model from these tests. Sausage Sizzle earns a *cost* model rather than a price model. The other seven are random-walk-like at the level of the price itself, which means any edge in them must come from somewhere other than the price's own history — from a related instrument, from the size of moves rather than their direction, or from a short-horizon overreaction.

Part III • Memory in direction

7. Autocorrelation — does yesterday's move predict today's?

IN PLAIN ENGLISH

The previous section asked about the price *level*. This one asks about the *moves*. If the price rose yesterday, does that tell us anything about today?

Three answers are possible. **Positive autocorrelation** means moves tend to continue — a rise makes another rise more likely. Traders call this momentum. **Negative autocorrelation** means moves tend to reverse — a rise makes a fall more likely. That is overreaction: the price jumped too far and is coming back. **Zero** means yesterday tells you nothing, which is the normal state of affairs in an efficient market.

Autocorrelation is just the correlation of the series with a shifted copy of itself. Do it at a shift of 1 day, 2 days, and so on, and you get a picture of how long the memory lasts.

THE MATHS

With daily changes r_t and their mean $\bar{r}$, the autocorrelation at lag k is

$$\rho(k) = \frac{\sum_{t=k+1}^n (r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=1}^n (r_t - \bar{r})^2}$$

Note this is exactly the IC formula from §2 with the signal set to "yesterday's move". Autocorrelation is an IC — it scores the simplest possible signal.

Under the null that the series is random, $\rho(k)$ is approximately Normal with standard deviation $1/\sqrt{n}$, giving the familiar **noise band of $\pm 1.96/\sqrt{n}$** drawn on every autocorrelation plot.

To test several lags at once rather than cherry-picking the best, the **Ljung-Box** statistic pools them:

$$Q(m) = n(n+2) \sum_{k=1}^m \frac{\rho(k)^2}{n-k} \sim \chi^2_m$$

A small p-value means "the first m lags contain more structure than chance allows", without our having had to choose m after seeing the data.

WORKED EXAMPLE — UQ DOLLAR'S P(1) BY HAND

Across $n = 364$ daily changes, mean change $\bar{r} = -0.00033$:

$$\sum (r_t - \bar{r})(r_{t-1} - \bar{r}) = -63.53 \quad \sum (r_t - \bar{r})^2 = 129.23$$

$$\rho(1) = -63.53 / 129.23 = \mathbf{-0.4916}$$

against a noise band of $\pm 1.96/\sqrt{364} = \pm 0.103$. The value is nearly five times the band. Practically, $\rho(1) = -0.49$ says that **about half of any day's move is undone the very next day**.

Autocorrelation of daily price changes — red = reverses, green = trends, grey = noise

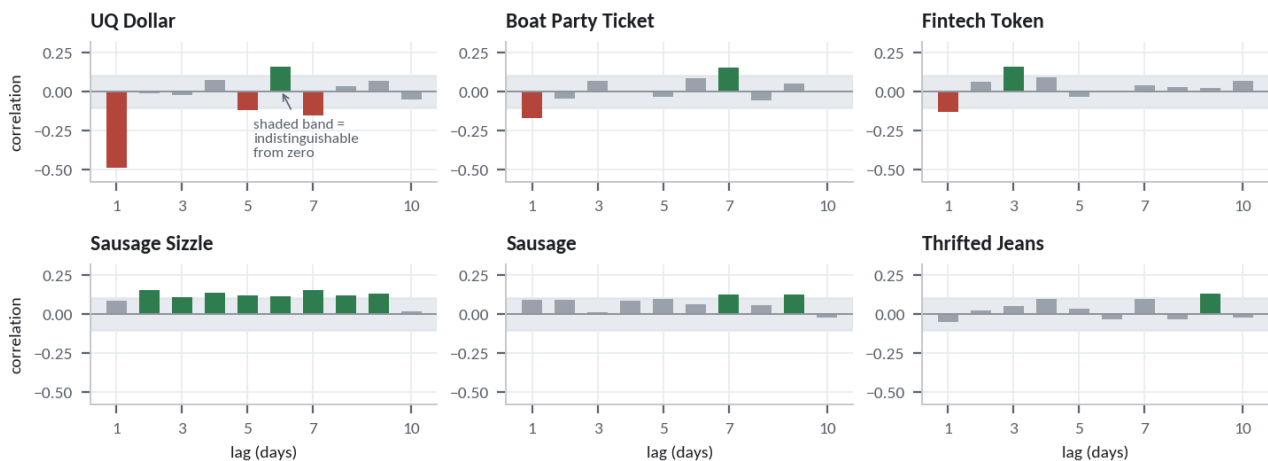


Figure 4. Autocorrelation of daily changes at lags 1 to 10 for six instruments. The grey band is the noise floor — bars inside it are indistinguishable from zero. Red bars below the band mean moves reverse; green bars above it mean moves continue. UQ Dollar's lag-1 bar is enormous and negative. Boat Party and Fintech Token show milder overreaction. Sausage Sizzle shows the opposite sign. Thrifted Jeans shows nothing at any lag.

8. Variance ratios — the same question at every horizon

IN PLAIN ENGLISH

Autocorrelation looks at one lag at a time, which makes it easy to miss a pattern that is spread thinly across many lags. The variance ratio test aggregates.

The idea reuses the drunkard's walk. If moves are independent, then the variance of a 10-day move should be exactly 10× the variance of a 1-day move — variances of independent things add up. So compute both and take the ratio. Getting 10 back means random. Getting less than 10 means the moves partly cancelled each other, which is reversion. More than 10 means they reinforced, which is trend.

Dividing through by the horizon makes the benchmark 1 instead of 10, which is how the statistic is conventionally reported.

THE MATHS

The **Lo-MacKinlay variance ratio** at horizon q :

$$VR(q) = \frac{(1/q) \cdot \text{Var}(q\text{-day change})}{\text{Var}(1\text{-day change})}$$

Estimated with the standard unbiased corrections, where $\hat{\mu} = \bar{r}$ and $m = q(n-q+1)(1 - q/n)$:

$$\hat{\sigma}_1^2 = \frac{1}{n-1} \sum_t (r_t - \hat{\mu})^2 \quad \hat{\sigma}_q^2 = \frac{1}{m} \sum_t \left(\sum_{j=0}^{q-1} r_{t-j} - q \hat{\mu} \right)^2$$

$$VR(q) = \hat{\sigma}_q^2 / \hat{\sigma}_1^2$$

The connection to §7 is exact and worth stating, because it shows VR is not a new idea but a weighted sum of the autocorrelations we already have:

$$VR(q) = 1 + 2 \sum_{k=1}^{q-1} (1 - k/q) \rho(k)$$

So VR simply adds up the autocorrelations with declining weights. Many small negative ρ 's that individually look like noise will together push VR clearly below 1.

WORKED EXAMPLE — UQ DOLLAR

$$\text{Var}(1\text{-day change}) = 0.3560 \quad \text{Var}(2\text{-day change}) = 0.1820$$

If moves were independent, the 2-day variance would be $2 \times 0.3560 = 0.7120$. It is 0.1820 — about **a quarter** of what randomness predicts. Dividing by the horizon:

$$VR(2) = 0.1820 / 0.3560 = \mathbf{0.511}$$

And it keeps falling as the horizon grows: $VR(5) = 0.214$, $VR(10) = 0.109$, $VR(20) = 0.060$. By 20 days the price has accumulated just 6% of the variance a random walk would have. It is going nowhere, extremely reliably.

9. Autocorrelation results

Instrument	$\rho(1)$	$\rho(2)$	Ljung-Box p (10 lags)	VR(10)	Reading
UQ Dollar	-0.492	-0.012	<0.001	0.109	Violent 1-day reversion
Boat Party	-0.168	-0.043	0.002	0.890	Overreaction
Fintech Token	-0.132	0.064	0.008	1.257	Weak 1-day reversion
Sausage Sizzle	0.085	0.155	<0.001	2.183	Trend (inherited from cost)
Sausage	0.091	0.090	0.002	1.752	Mild trend
Bread	0.064	0.031	0.234	1.560	Mild trend
MenuDash	-0.042	-0.027	0.362	0.632	Reversion in level, not in moves
Thrifty Jeans	-0.051	0.023	0.077	1.266	Nothing
Liferaft	0.039	-0.026	0.215	0.992	Nothing

The two methods agree, which is reassuring: everything with a negative $\rho(1)$ also has VR below 1, and everything with a positive $\rho(1)$ has VR above 1.

MenuDash is the interesting disagreement. Its $\rho(1)$ is inside the noise band, so day-to-day it looks random — yet $VR(10) = 0.63$ says that over ten days its moves substantially cancel. That is the signature of a *level* that reverts while the individual steps look unremarkable, and it is exactly what §13 finds: MenuDash oscillates around a fair value it can't stray far from.

A caution we should state plainly: the formal z-statistics on these variance ratios are all small ($|z| < 0.5$), because the Lo-MacKinlay test has poor power on 365 observations. We use VR as a corroborating indicator of direction, not as standalone proof. The split-half IC test in §3 is what actually decides.

Part IV • Memory in size

10. Volatility clustering — the calm and the storm

IN PLAIN ENGLISH

There is a second kind of memory in prices, and it is usually much stronger than the first: **the memory of how big moves are, regardless of direction.**

Anyone who has watched a market has noticed that turbulence comes in spells. Quiet weeks follow quiet weeks; once things get violent they stay violent for a while. The direction of tomorrow's move may be a coin flip, but *the size* of it is quite predictable from today.

That matters for two reasons. First, position sizing: taking the same number of units in a calm regime and a wild one means taking wildly different amounts of risk. Second, signal scaling: a \$10 deviation means something very different when typical daily moves are \$7 versus \$20, so dividing a signal by current volatility makes it comparable across regimes.

The way to test it is a trick. To study size and ignore direction, throw away the sign — square the moves, or take absolute values — and then run the same autocorrelation machinery on those.

THE MATHS

The ARCH-LM test. Regress squared changes on their own recent past:

$$r_t^2 = \gamma_0 + \gamma_1 r_{t-1}^2 + \dots + \gamma_5 r_{t-5}^2 + u_t$$

Under the null of no clustering, all γ are zero and the statistic $n \cdot R^2$ follows χ^2_5 . A small p-value means volatility genuinely clusters.

The GARCH(1,1) model. Having established clustering exists, GARCH describes it. Write σ_t^2 for the variance we expect on day t :

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

ω — a floor; the long-run variance the process returns to.

α — how sharply today's variance reacts to yesterday's actual shock.

β — how much of yesterday's *expected* variance carries over.

Read it as a forecast that updates: your new estimate of volatility is a blend of a long-run anchor, the surprise you just observed, and your previous estimate. α controls how jumpy the forecast is, β how sticky.

Persistence. The quantity $\alpha + \beta$ governs how long a volatility shock survives. By the same argument used for the OU half-life, a shock decays by a factor $(\alpha + \beta)$ per day, so:

$$\text{volatility half-life} = \frac{\ln 1/2}{\ln(\alpha + \beta)}$$

Values near 1 mean shocks last for months. Above 1 the process is explosive and the model is misspecified.

WORKED EXAMPLE — FINTECH TOKEN

Fitting GARCH(1,1) gives $\alpha = 0.035$, $\beta = 0.951$, so persistence is $\alpha + \beta = \mathbf{0.986}$ and

$$\text{half-life} = \ln 0.5 / \ln 0.9855 = \mathbf{47.5 \text{ days}}$$

A volatility shock takes about seven weeks to half-fade. We can see it directly in the data: the calmest 20-day stretch (around day 323) has a standard deviation of \$7.41, the wildest (around day 133) \$19.66 — a **2.65× range** — and that rolling volatility series has an autocorrelation of **0.967** from one day to the next. It moves like a slow tide.

Meanwhile the |move| autocorrelation is still positive at lag 10 (0.125), which is long memory: how big today's move was still tells you something about a move two weeks away.

Fintech Token: direction is unpredictable, size is not

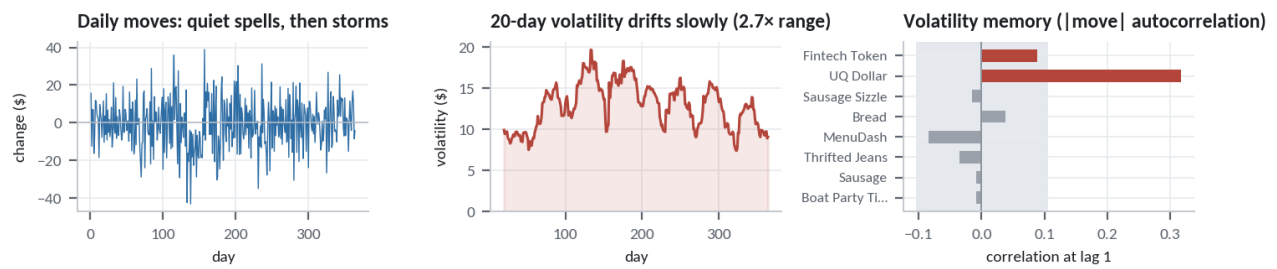


Figure 5. Left: Fintech Token's daily changes, where the alternating calm and turbulent spells are visible without any statistics. Middle: 20-day rolling volatility, drifting slowly across a 2.7× range rather than jumping around. Right: lag-1 autocorrelation of |move| across the board — only Fintech Token and UQ Dollar clear the noise band.

Where this test can mislead you, and did. Four instruments come back with GARCH persistence above 0.97 — Sausage Sizzle 0.979, Bread 0.979, MenuDash 0.984. Taken at face value that suggests strong volatility clustering everywhere.

It is an artefact. Their ARCH-LM p-values are 0.30, 0.64 and 0.34 — no evidence of clustering at all. What is happening is that GARCH is a flexible model being handed a series with a slow upward trend in its *level*, and it fits that trend by reporting high persistence. The formal test says there is nothing there.

When a model output and a hypothesis test disagree, believe the test. Only Fintech Token ($p = 0.004$) and UQ Dollar ($p < 0.001$) have real ARCH effects.

Instrument	ARCH-LM p	α	β	$\alpha+\beta$	Vol half-life	Real?
Fintech Token	0.004	0.035	0.951	0.986	47.5d	yes
UQ Dollar	<0.001	0.463	0.078	0.541	1.1d	yes
Sausage Sizzle	0.30	0.006	0.973	0.979	32.5d	no
Bread	0.64	0.015	0.964	0.979	33.1d	no
MenuDash	0.34	0.000	0.984	0.984	41.8d	no
Thrifty Jeans	0.82	0.000	0.899	0.899	6.5d	no
Sausage, Liferaft, Boat Party	>0.5	~0	~0.50	~0.50	~1d	no

UQ Dollar's profile is worth a sentence because it is the opposite of Fintech's. Its $\alpha = 0.46$ with $\beta = 0.08$ means volatility reacts violently to yesterday's shock but forgets it immediately — one-day memory, no persistence. That is not clustering in the usual sense; it is just the mechanical consequence of the spring. A big deviation today *forces* a big correcting move tomorrow.

WHAT IT MEANS FOR TRADING

Use volatility persistence for **sizing, not direction**. Fintech Token's volatility is forecastable seven weeks out while its direction is barely forecastable one day out. Concretely, dividing the reversion signal by a rolling 20-day volatility estimate lifts its IC from 0.133 to **0.151** — a free improvement from the same underlying trade, because it stops us taking oversized positions during storms.

Part V • Relationships between instruments

11. Lead-lag: does one instrument move before another?

IN PLAIN ENGLISH

So far every test has looked at each instrument in isolation. But an edge can also live *between* instruments: if A reliably moves before B, then watching A tells us what B will do.

The test is simple correlation again, with a shift. Line up A's move on Monday against B's move on Tuesday, and across the whole year, measure how well they agree. Repeat for every ordered pair and every lag from 1 to 5 days.

One warning before the results. With 9 instruments there are 72 ordered pairs, and 5 lags each — 360 correlations. At a 5% significance threshold, roughly 18 will look significant *by pure chance*. So we should only take seriously the ones that stand far above the crowd, and we should be suspicious of any result that doesn't come with a story about *why* the relationship would exist.

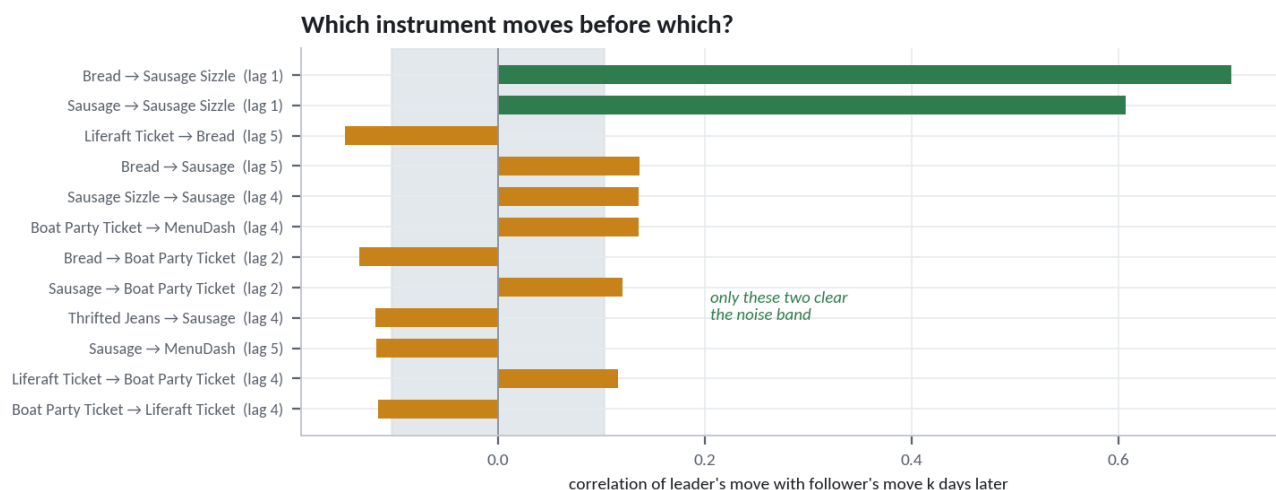


Figure 6. The twelve strongest lead-lag correlations out of 360 tested. Two tower over the rest: Bread and Sausage each predict Sausage Sizzle one day later, at 0.71 and 0.61. Everything below them sits at or barely above the noise band and is consistent with what 360 random tests would produce anyway.

Same-day correlations, incidentally, are all below 0.13 in absolute value — the instruments are essentially independent within a day. **All the cross-instrument structure is in the lags.**

To confirm the two survivors aren't a fluke we ran **Granger causality**, which asks whether adding A's past to a model of B improves the forecast beyond B's own past. Bread → Sizzle and Sausage → Sizzle both come back at $p < 0.001$. The next best pair on the board is $p = 0.031$, which is exactly what 72 tests at the 5% level would throw up by accident.

12. Sausage Sizzle — tomorrow's price is on today's receipt

IN PLAIN ENGLISH

The lead-lag result has an obvious real-world reading, and it is the single most valuable thing in this dataset.

A sausage sizzle is made of bread and sausages, plus labour. If the stall buys ingredients one day and sells the next, then the price it charges is determined by *yesterday's* ingredient prices. So:

Today we can already see what bread and sausages cost. Those are printed numbers, not forecasts. If the sizzle price is a formula applied to them, then **tomorrow's sizzle price is not a prediction at all — it is an arithmetic problem we can solve today.**

That is not a statistical edge in the usual sense. It's closer to reading the answer sheet.

THE MATHS

Fit by ordinary least squares:

$$\text{Sizzle}_t = c + A \cdot \text{Bread}_{t-1} + B \cdot \text{Sausage}_{t-1} + \varepsilon_t$$

A, B — how much bread and sausage go into one sizzle.

c — the fixed component, which we interpret as the labour cost.

ε_t — whatever the recipe doesn't explain.

The forecast we actually trade is the *change*, and it needs no forecasting at all because both terms are known on day t :

$$\Delta \text{Sizzle}_{t+1} \approx A \cdot \Delta \text{Bread}_t + B \cdot \Delta \text{Sausage}_t + (\text{drift in labour})$$

To make the labour term concrete: rearranging the first equation gives the implied labour cost on each day as the residual,

$$\text{Labour}_t = \text{Sizzle}_t - A \cdot \text{Bread}_{t-1} - B \cdot \text{Sausage}_{t-1}$$

and its recent slope is our estimate of the drift.

WORKED EXAMPLE

Fitting on all 364 usable days:

$$\text{Sizzle}_t = 18.9495 + 0.07422 \cdot \text{Bread}_{t-1} + 1.72683 \cdot \text{Sausage}_{t-1}$$

with $R^2 = 0.696$ and a residual standard deviation of \$0.66.

Sanity-check the coefficients against reality: 1.73 sausages and 0.074 of a loaf per serve. A loaf makes roughly 13 slices, so 0.074 of a loaf is about one slice. **One slice of bread and roughly two sausages** — the regression has recovered a plausible recipe from nothing but price data. (For reference, `algorithm.py` currently uses 0.07692 and 1.76921, within 4% of these.)

The residual — the implied labour cost — averages \$18.95 with a standard deviation of \$0.66, but its **day-to-day change has a standard deviation of only \$0.0375**. It barely moves. That is why the change equation works so well: nearly everything that moves the sizzle price tomorrow is visible today.

Watch it work on eight consecutive days:

Day	Bread move	Sausage move	Predicted sizzle move	Actual
100	-0.01	0.00	-0.001	-0.05
101	+1.33	0.00	+0.099	+0.07
102	-1.10	+0.01	-0.064	-0.11
103	-0.14	0.00	-0.010	0.00
104	+0.47	-0.01	+0.018	-0.05
105	+0.54	+0.02	+0.075	+0.09
106	+0.64	+0.05	+0.134	+0.09
107	-1.00	0.00	-0.074	-0.11

Six of eight directions correct; the magnitudes are close. Across the full year:

$$\text{IC} = \mathbf{0.922} \quad R^2 = 0.850 \quad \text{sign correct on } \mathbf{83.2\%} \text{ of days}$$

Adding a 10-day labour drift term lifts the IC to **0.960**, and the split halves are 0.966 and 0.953 — as stable as anything gets.

Sausage Sizzle: tomorrow's price is already on today's receipt



Figure 7. Left: the recipe's prediction laid over the actual sizzle price. Middle: the left-over labour cost, which wanders slowly and never jumps — the reason the forecast holds up. Right: today's known ingredient cost change against tomorrow's actual sizzle move. Points sit close to the line; this is what an IC of 0.92 looks like.

Why $R^2 = 0.70$ but $IC = 0.92$? These describe different problems and the gap is instructive. The R^2 of 0.70 is for predicting the *level*, where the model must also track the slow labour drift it can't see. The IC of 0.92 is for predicting the *change*, where the drift almost entirely cancels out because it barely moves day to day. Predicting where something is going is often much easier than predicting where it is.

WHAT IT MEANS FOR TRADING

Do not trade Sausage Sizzle with momentum. `algorithm.py`'s current 13-day momentum signal scores an IC of 0.263 — respectable, and roughly a quarter as good as simply computing the answer. Replace it with the recipe forecast and take the full position on its sign.

For Round 2, re-fit *A* and *B* online: a rolling OLS of Sizzle on lagged Bread/Sausage reaches $R^2 = 0.70$ immediately and needs no tuning. The current smoothness-minimisation refit is a reasonable idea but a more roundabout route to the same coefficients.

13. MenuDash — the same cost, seen through frosted glass

IN PLAIN ENGLISH

The sizzle regression left us with a by-product: a day-by-day estimate of the hidden labour cost. It turns out MenuDash is tracking that same cost — but imprecisely. It rounds to the nearest cent, it adds noise, and it lags.

That sets up a classic trade. We have a *precise* measurement of something (from the sizzle) and a *noisy* one (MenuDash). When the noisy one wanders away from what the precise one implies, we bet it comes back.

The important discipline here is that the fair value must be estimated using only past data. It would be easy — and wrong — to fit the relationship over the whole year and then measure how well the resulting gap predicts. We use a rolling 60-day window that only ever looks backwards.

THE MATHS

On each day t , using only days $t-60 \dots t-1$, fit

$$\text{MenuDash}_s = p + q \cdot \text{Labour}_s + \varepsilon_s$$

then form today's fair value and the gap to trade:

$$\text{Fair}_t = \hat{p} + \hat{q} \cdot \text{Labour}_t \quad \text{Gap}_t = \text{Fair}_t - \text{MenuDash}_t$$

A positive gap means MenuDash is cheap relative to the cost it tracks, so we buy.

WORKED EXAMPLE

Over the full year, MenuDash's level correlates **0.931** with the implied labour cost ($R^2 = 0.867$). Fitting the gap's own reversion the same way we did for UQ Dollar in \$4 gives a **half-life of 2.94 days** — mispricings close within about three days.

The causal, rolling version of this signal tests at **IC = 0.280** ($t = 4.90$), with halves of 0.301 and 0.265. Compare that to a naive 9-day mean-reversion signal on MenuDash alone, which manages only 0.135. Knowing what the price *should* be is worth roughly double.

MenuDash: a blurry, rounded readout of the same cost



Figure 8. Left: the first 120 days, drawn as steps to show the price only ever moves in whole cents — and on 22.5% of days doesn't move at all. Middle: MenuDash against the labour cost implied by the sizzle recipe; they track closely. Right: the gap between price and fair value, which oscillates around zero and closes in about three days.

WHAT IT MEANS FOR TRADING

The current 60-day rolling regression is the right approach — keep it. But add a **deadband of about one tick (\$0.01)**. MenuDash trades on a one-cent grid and doesn't move at all on 22.5% of days, so acting on a gap smaller than a tick generates turnover without generating profit. In the naive backtest MenuDash churned 12.7 million units, by a wide margin the worst on the board.

14. Cointegration — pairs that can't drift apart

IN PLAIN ENGLISH

Two prices can each wander like a balloon, yet be tied to each other by a string. Individually unpredictable; the *distance between them* mean-reverting. That is cointegration, and it underlies most pairs trading.

We tested all 36 pairs. The results are a good lesson in not believing statistics without a mechanism.

THE MATHS

The Engle-Granger procedure: regress one on the other to find the hedge ratio, then test whether what's left over is stationary.

$$Y_t = a + b X_t + z_t \rightarrow \text{run the ADF test of } \$4 \text{ on the spread } z_t$$

If the spread is stationary, the pair is cointegrated and b tells you how many units of X to hold against one of Y .

WORKED EXAMPLE — THREE FALSE POSITIVES AND TWO COINCIDENCES

The top of the results table looks spectacular:

Pair	p-value	Spread half-life
UQ Dollar ~ Bread	<0.001	0.69d
UQ Dollar ~ Sausage Sizzle	<0.001	0.71d
UQ Dollar ~ Sausage	<0.001	0.71d
Thrifty Jeans ~ Sausage Sizzle	0.0013	17.5d
Fintech Token ~ Bread	0.0091	18.1d

The first three are **not real**. UQ Dollar is stationary all by itself, so any combination of it with anything else inherits that stationarity. The test is detecting UQ Dollar's own spring, not a relationship. Notice the half-lives — 0.7 days, exactly UQ Dollar's own — which gives the game away.

The next two are statistically respectable but we reject them anyway. We tested 36 pairs; at a 5% threshold we'd expect about two hits by chance, and we got about two. Neither has any story behind it — there is no reason thrifty jeans should be tied to a sausage sizzle. Under a Bonferroni correction for 36 tests the threshold becomes $0.05/36 = 0.0014$, and only the Jeans-Sizzle pair scrapes through.

WHAT IT MEANS FOR TRADING

No pairs trades. The only genuine cross-instrument relationship on this board is the sizzle recipe, and that is better traded directly as a forecast than as a spread.

Part VI • Cross-sectional mean reversion

15. Do today's winners become tomorrow's losers?

IN PLAIN ENGLISH

This is a different shape of question from everything so far. Instead of asking whether an instrument reverts against *its own* history, we ask whether it reverts against *the others*.

The idea: each day, rank all nine by how they performed relative to the group. If the best performer tends to underperform tomorrow and the worst tends to bounce, you can trade it by shorting the winners and buying the losers — no view on the market as a whole required. This is a real and well-documented effect in equity markets.

Testing it takes one preparatory step. The instruments have wildly different volatilities — Thrifted Jeans swings at 86% a year, Sausage Sizzle at 4% — so a raw comparison would be dominated by the volatile names. We first divide each instrument's returns by its own volatility to put them on a common scale, then subtract the daily average across the nine so we're left with purely relative performance.

THE MATHS

For instrument i on day t , with percentage return $r_{i,t}$:

$$z_{i,t} = \frac{r_{i,t} - \mu_i}{\sigma_i} \quad (\text{standardise each instrument}) \quad x_{i,t} = z_{i,t} - \frac{1}{N} \sum_j z_{j,t} \quad (\text{remove the market-wide move})$$

Then pool every instrument-day into one regression and measure

$$\text{corr}(x_{i,t-1}, x_{i,t})$$

A negative value means relative winners become relative losers: cross-sectional mean reversion.

WORKED EXAMPLE — AND WHY WE THREW IT OUT

The headline result is compelling:

$$\text{corr} = -0.093, \quad t = -5.32, \quad n = 3,267 \text{ instrument-days}$$

A t-statistic of -5.3 is far past any conventional threshold. Taken at face value, we have found a cross-sectional reversal factor.

But a pooled statistic can be dominated by one member of the pool. UQ Dollar's own autocorrelation is -0.49 ; when we drop it into a panel, that reversion shows up in the panel average whether or not any *relative* effect exists. So we re-ran the test on subsets:

Panel	corr	t	Significant?
All 9 instruments	-0.093	-5.32	yes
Drop UQ Dollar	-0.042	-2.26	marginal
Drop UQ Dollar + Liferaft	-0.066	-3.31	yes
Drop all four with their own structure	-0.023	-0.98	no

Removing the four instruments that already have documented reversion of their own — UQ Dollar, Liferaft, Boat Party and Sizzle — leaves five, and among those five the effect is **-0.023 with $t = -0.98$** . Nothing.

The per-instrument breakdown confirms it. After removing the market-wide move, the own-reversion values are UQ Dollar -0.430 , Boat Party -0.148 , Sizzle -0.126 , Fintech -0.103 — and then Thrifted Jeans -0.025 , Bread -0.014 , Sausage -0.008 . The effect lives entirely in instruments we already knew reverted.

Cross-sectional mean reversion: a mirage

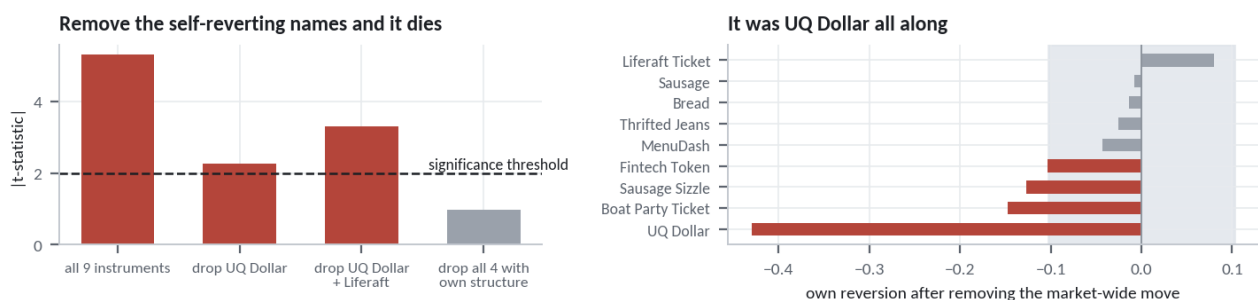


Figure 9. Left: significance of the cross-sectional reversal effect as instruments are removed. It survives while the self-reverting names are in the panel and dies when they're taken out. (It isn't monotone — dropping Liferaft, which reverts *positively*, temporarily strengthens the measured negative — but the final bar is the one that matters.) Right: each instrument's own reversion after removing the market-wide move. Four red bars, five at zero.

WHAT IT MEANS FOR TRADING

Do not build a rank-and-reverse book. There is no cross-sectional factor here; there are four instruments with individual mean reversion, which are already better traded on their own terms.

There is a silver lining. The same near-zero correlations that kill the cross-sectional trade mean the instruments are close to independent — so a portfolio of the validated single-instrument signals gets genuine diversification for free, with no correlated drawdowns to manage.

Part VII • Seasonality, jumps, and a trap we fell into

16. Looking for hidden cycles

IN PLAIN ENGLISH

Prices sometimes follow a calendar — weekly patterns, monthly settlement effects, semester cycles at a university. The tool for finding cycles you can't see is the **periodogram**: it decomposes a wiggly series into pure waves of different periods and reports how much of the total movement each wave accounts for.

If there were a strong weekly rhythm, a wave with a 7-day period would light up.

THE MATHS

Take the discrete Fourier transform of the de-meaned series $x_t = P_t - \bar{P}$:

$$X(f) = \sum_{t=0}^{n-1} x_t e^{-2\pi i f t} \quad S(f) = \frac{|X(f)|^2}{n}$$

$S(f)$ is the power at frequency f ; the corresponding period is $1/f$ days. A genuine cycle produces a sharp spike.

As a cross-check we also ran a one-way ANOVA, grouping days by their position in a 5-, 7- and 30-day cycle and testing whether the group means differ.

WORKED EXAMPLE

For eight of the nine instruments, the largest peak sits at a period of 365 or 182.5 days — which is the length of the sample itself, or half of it. **That is not a cycle; it is the trend.** Any series that ends somewhere different from where it started will show power at the sample length.

The ANOVA agrees. Testing periods 5, 7 and 30 across all nine instruments gives 27 p-values, of which two fall below 0.05 — precisely what you'd expect by chance.

No weekly or monthly seasonality exists anywhere on this board.

The exception is Boat Party Ticket, where 50% of the power sits at period 182.5. With two semesters in a 365-day year, that is a real calendar effect. It is also, as the next section shows, harder to use than it looks.

17. The Boat Party trap — a lookahead error, and how we caught it

IN PLAIN ENGLISH

Boat Party Ticket has two visible humps corresponding to two semesters. 83.8% of its variance is that smooth seasonal shape and only 7.7% is short-term wiggle. That is enormous, and the obvious idea is to trade the shape: work out where in the calendar we are and lean the position that way.

Our first attempt measured the seasonal slope using a **centred** 21-day moving average — the average of the 10 days before and the 10 days after. It scored an IC of 0.258, the best result on the board outside the top two.

It was wrong. A centred average on day t includes days $t+1$ through $t+10$. We were using next week's prices to predict tomorrow. On the real trading day that information does not exist.

WORKED EXAMPLE — THE SIZE OF THE ERROR

Redoing it honestly with a *trailing* average that only looks backwards:

Signal	IC	t	Halves	Valid?
Centred 21-day slope	0.258	4.94	0.29 / 0.21	no — lookahead
Trailing 10-day slope	0.029	0.53	0.05 / 0.00	yes, but worthless
Trailing 21-day slope	-0.022	-0.40	0.05 / -0.12	yes, but worthless
Plain 3-day reversion	0.181	3.50	0.09 / 0.30	yes

The lookahead was inflating a worthless signal (0.029) into an apparently strong one (0.258).

That still leaves the hard-coded seasonal table in `algorithm.py`, which is fitted on Round 1 and applied to Round 2 — legitimate in principle, since it uses no future data within the round. So we tested whether such a table transfers at all: fit the day-of-year shape on days 0-181, apply it to days 182-364.

$$\text{IC} = -0.118 \quad (t = -1.53, n = 167)$$

It points the wrong way. Round 1's two semesters are not copies of each other — Figure 10 shows them overlaid, and the peaks land in different places. A fixed calendar table has no demonstrated ability to transfer.

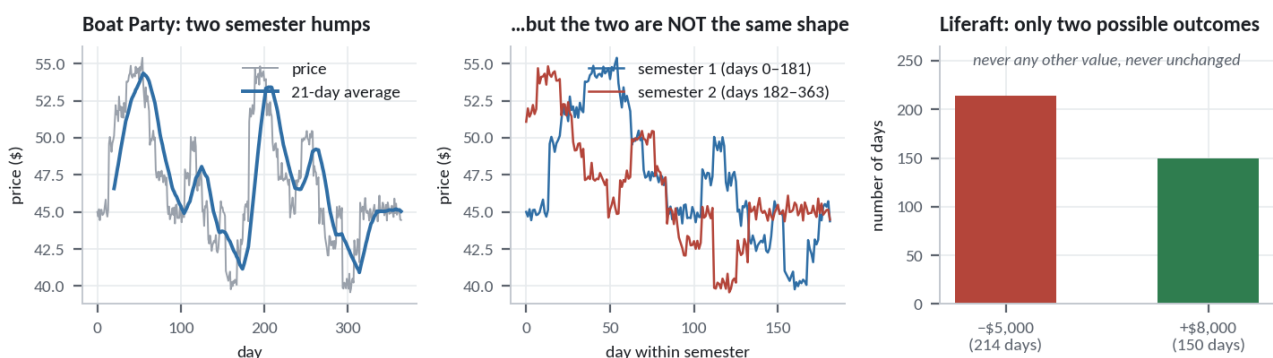


Figure 10. Left: Boat Party's two seasonal humps. Middle: the two semesters drawn on top of each other — visibly different shapes, which is why a fixed day-of-year table fitted on one does not help on the other. Right: Liferaft's daily changes take exactly two values and nothing else.

WHAT IT MEANS FOR TRADING

Drop the `BOAT_SEASONAL` table, or heavily de-weight it. Trade the part that validates: plain 3-day mean reversion, IC 0.181, stable across halves. If you want seasonal exposure, use a *causal* trailing-21-day slope which at least adapts to whatever Round 2's calendar turns out to be — the combination tests at 0.206, slightly better than reversion alone.

Boat Party is also the best instrument per dollar of budget consumed (1.998), so it deserves its high position in the budget priority list.

18. Jumps and fat tails

Only Boat Party has genuine jumps: 7 moves beyond 3 standard deviations and 3 beyond 4. Its **excess kurtosis is 9.5**, where a Normal distribution scores 0. (Reported as raw kurtosis the figures are 12.5 against a Normal's 3 — the two conventions differ by exactly 3, and mixing them up is a common slip.) Extreme days are far more frequent here than a bell curve allows. Every other instrument passes a Jarque-Bera normality test.

What happens after a jump is asymmetric. The average next-day move is **+0.243 after a big drop** and **+0.115 after a big rise** — both positive. The snap-back is real but mostly follows crashes, not spikes.

19. Liferaft Ticket — not a price at all

IN PLAIN ENGLISH

Liferaft is the one instrument whose price isn't generated in advance. It is computed from what every competing team decides: if most of the room goes long, it falls; if most goes short, it rises. It is a voting game wearing a price's clothing.

The data makes this unmistakable. Its daily change takes **exactly two values**, $-\$5,000$ on 214 days and $+\$8,000$ on 150 days. Never anything in between, never zero. That is not something a price process does.

The consequence matters more than the curiosity: Round 1's path is a record of what Round 1's room did. Round 2 has a different room. **There is nothing here to learn from.**

THE MATHS

Let $p = P(\text{the majority goes long})$. Then per day:

$$E[\text{long}] = 8000 \cdot (1 - p) - 5000 \cdot p \quad E[\text{short}] = 5000 \cdot p - 8000 \cdot (1 - p) = -E[\text{long}]$$

Long beats short whenever $E[\text{long}] > 0$:

$$8000(1 - p) > 5000p \implies p < \frac{8000}{8000 + 5000} = \mathbf{0.615}$$

The payoff asymmetry does the work: winning long pays \$8,000 while winning short pays only \$5,000, so long is favoured unless the room is heavily long.

WORKED EXAMPLE

Round 1's observed rate was $214 / 364 = \mathbf{0.588}$, below the 0.615 breakeven. At that rate:

$$E[\text{long}] = 8000 \times 0.412 - 5000 \times 0.588 = \mathbf{+\$357 \text{ per day}}$$

We also tested whether the room's behaviour is predictable — whether a run of short days predicts another. The autocorrelation of the move sign is 0.039 against a noise band of 0.103, and the mean run length is 2.14, both consistent with independent coin flips. The contrarian rule in `algorithm.py` scores an IC of $\mathbf{-0.039}$ ($t = -0.74$).

WHAT IT MEANS FOR TRADING

Hold a constant long and ignore the price history. The $+\$8,000 / -\$5,000$ asymmetry favours long unless more than 61.5% of teams go long, and Round 1 suggests they won't. The current adaptive contrarian logic is fitting noise; its shrinkage toward an even room makes it mostly harmless, but a constant long is at least as good and cannot be whipsawed by a random streak.

Treat 0.588 as a weak prior, not a signal — and be aware that other teams reading the same specification may reach the same conclusion, which would push the room long and erode the edge.

Part VIII • Models and recommendations

20. Everything that survived

Instrument	Model	IC	t	1st half	2nd half
Sausage Sizzle	Known ingredient move + 10-day labour drift	0.960	64.3	0.966	0.953
UQ Dollar	OU reversion to fitted peg of 99.94	0.694	18.3	0.726	0.680
MenuDash	Rolling causal fair-value gap to labour	0.280	4.90	0.301	0.265
Boat Party	3-day reversion + causal trailing slope	0.206	3.88	0.141	0.290
Sausage	11-day momentum	0.180	3.43	0.262	0.073
Bread	10-day momentum	0.136	2.58	0.132	0.136
Fintech Token	2-day reversion, volatility-scaled	0.133	2.54	0.059	0.243

21. Everything that didn't

Instrument	Model tested	Result	Why rejected
Thrifty Jeans	15-day momentum	IC 0.097	t = 1.82, below threshold
Fintech Token	5-, 7-, 10-day reversion	IC -0.016 to -0.062	sign flips between halves
Liferaft	Contrarian to last move	IC -0.039	t = -0.74
Boat Party	Fixed day-of-year table	IC -0.118 out of sample	doesn't transfer
All	Cross-sectional reversal	IC -0.023 ex-UQD	t = -0.98
All pairs	Cointegration trades	2 of 36 at p<0.05	expected by chance

22. Per-instrument recommendations

Sausage Sizzle — a forecast, not a signal. Compute $\Delta \text{Sizzle}_{t+1} = 0.0742 \cdot \Delta \text{Bread}_t + 1.7268 \cdot \Delta \text{Sausage}_t + \text{drift}(\text{labour}, 10\text{d})$ and take the full position on its sign. Re-fit the coefficients online with a rolling OLS. This is the highest-conviction trade available.

UQ Dollar — Ornstein-Uhlenbeck, sized by the gap. $\kappa = 0.963$, half-life 0.72 days, peg \$99.94, residual $\sigma = \$0.43$. The current implementation uses a binary signal with a \$0.15 deadband; the bucket analysis in Figure 3 shows hit rate rising smoothly from 49% at tiny gaps to 92–95% at large ones. **Size proportionally:** position $\propto \text{clip}(\text{deviation} / 0.5, -1, +1)$. The current deadband discards the \$0.05–\$0.15 bucket, which still hits 59–72%.

MenuDash — rolling cointegration to the labour cost. Keep the 60-day rolling regression. Add a one-tick (\$0.01) deadband to stop churning on sub-tick gaps.

Boat Party — short-horizon reversion only. 3-day mean reversion. Drop or de-weight the hard-coded seasonal table (§17). Best instrument per dollar of budget, so keep it high in the priority list.

Fintech Token — volatility-scaled 1-day reversion. Replace the current 5/7/10 window average, all of which fail split-half, with a 1–2 day reversion divided by a rolling-20-day volatility estimate. The volatility persistence ($\alpha + \beta = 0.986$) belongs in the sizing, not the direction.

Bread and Sausage — keep momentum, fund last. Both are stable, and the edge is genuine time-series momentum rather than drift capture (drift-removed ICs are identical to raw). But they are the two worst instruments per dollar of budget (0.301 and 0.399), and Sausage turned over 485,000 units in the naive backtest.

Thrifty Jeans — hold the partial long, and know it's a bet. The momentum signal doesn't validate and one year cannot establish the drift. The current 0.6 partial long is a defensible compromise, but at 86% annualised volatility a 119% Round 1 gain could just as easily be -50% in Round 2.

Liferaft — constant long. See §19.

23. The budget constraint

Taking every position at its full limit simultaneously would require **\$693,066** against a **\$600,000** cap, so the priority ordering genuinely binds — something has to be cut every day.

Ranked by profit per dollar of budget consumed in the naive backtest:

Rank	Instrument	Gross P&L	Sharpe	Avg notional	P&L per \$
1	Boat Party Ticket	\$92,610	4.13	\$46,362	2.00
2	Thrifty Jeans	\$77,512	1.58	\$43,263	1.79
3	Fintech Token	\$88,422	2.96	\$59,767	1.48
4	UQ Dollar	\$56,986	7.09	\$62,111	0.92
5	Liferaft Ticket	\$100,000	0.69	\$147,381	0.68
6	Sausage	\$10,650	2.80	\$26,709	0.40
7	Bread	\$18,485	1.89	\$61,377	0.30
8	MenuDash	\$40,500	1.27	\$136,482	0.30
9	Sausage Sizzle	\$28,380	4.46	\$109,615	0.26

Read this table with care — the ranking is not the whole story. Sausage Sizzle comes last on profit per dollar, yet it carries by far the most reliable signal on the board (IC 0.96 against 0.13–0.28 for everything else). Its low ratio reflects that the sizzle simply doesn't move much, not that the trade is bad.

Two considerations pull in opposite directions. Raw P&L per dollar says fund it last. Certainty says fund it first, because a near-certain small gain is worth more than an uncertain larger one once you weight by risk. Note that Sizzle and UQ Dollar have the two highest Sharpe ratios among the reliable signals (4.46 and 7.09) — Sharpe captures exactly this, and it tells a different story from the raw ratio.

We would promote Sausage Sizzle above its current sixth place. But this is a judgement call about risk appetite rather than something the data settles, and it deserves a dedicated backtest under the real budget constraint rather than the unconstrained figures above.

Appendix A • Caveats

One year is not much data. Everything here rests on 365 observations. A t -statistic of 2.5 on 350 points is real but not strong; only the top three findings (Sizzle, UQ Dollar, MenuDash) have the kind of margin that would survive serious scrutiny. Where a result is marginal we have said so.

We tested a lot of things. Roughly 36 cointegration pairs, 360 lead-lag correlations, 72 Granger pairs, and dozens of signal windows per instrument. We have not applied a formal family-wise correction throughout; instead we required split-half stability, which serves a similar purpose more practically. Readers should still treat any single marginal result with suspicion.

The ICs are gross of the budget constraint. They assume the full limit is taken on the sign of the signal every day. Real P&L will be lower once nine positions compete for \$600,000.

Structure should transfer; constants may not. Round 2 is a fresh draw from the same generators. The *mechanisms* — the peg, the recipe, the rounding, the voting game — should still be there. The *fitted numbers* — drift rates, the seasonal shape, the P&L rankings — may not be. Prefer models that re-fit online.

Liferaft is adversarial. Its payoff depends on what other teams do. Any edge there erodes as more teams work out the same asymmetry.

Appendix B • Glossary

Term	Meaning
ACF (autocorrelation function)	Correlation of a series with a delayed copy of itself. Measures memory.
ADF test	Augmented Dickey–Fuller. Tests whether a series mean-reverts or wanders freely.
ARCH-LM	A test for whether the size of moves clusters in time.
Cointegration	Two series that individually wander but whose difference mean-reverts.
Drift	A persistent average tendency to move in one direction.
GARCH(1,1)	A model of how volatility evolves: a blend of a long-run anchor, yesterday's shock and yesterday's forecast.
Granger causality	Tests whether A's past improves a forecast of B beyond B's own past. Not causality in the everyday sense.
Half-life	Days needed to close half of a gap. The natural measure of reversion speed.
Hurst exponent	How fast a series travels as the horizon grows. 0.5 = random, below = reverting, above = trending.
IC (Information Coefficient)	Correlation between a signal and the next period's move. Our scoring metric throughout.
Jarque–Bera	Tests whether a distribution has Normal skewness and kurtosis.
Kurtosis	Fatness of the tails. Normal = 3; higher means more extreme events.
Ljung–Box	Tests several autocorrelation lags jointly instead of one at a time.
Lookahead bias	Accidentally using information that wouldn't have been available. The most common way backtests lie.
Mean reversion	The tendency to be pulled back toward a level.
Momentum	The tendency for moves to continue.
Ornstein–Uhlenbeck	The standard mean-reverting process; a spring with random shocks.
Periodogram	Decomposes a series into waves to reveal hidden cycles.
Sharpe ratio	Return divided by volatility, annualised. Reward per unit of risk.
Stationary	Having a stable mean and variance over time. A stationary price has a home.
t-statistic	How many standard errors an estimate sits from zero. Above 2 is the usual bar.
Variance ratio	Compares long-horizon to short-horizon variance. Below 1 = reversion, above 1 = trend.

Appendix C • Reproducing this analysis

All results are generated from the nine CSVs in `trader_interface/data/`.

Script	What it does
<code>structure_tests.py</code>	Stationarity, Hurst, distributions, ACF/PACF, Ljung–Box, variance ratios, ARCH-LM, GARCH, lead-lag, Granger, cointegration, cross-sectional reversion, periodogram. Writes <code>out/*.csv</code> .
<code>deep_dive.py</code>	Per-instrument follow-ups: OU fit, recipe regression, MenuDash tick structure, Boat Party decomposition, Fintech regimes, drift-vs-momentum, Liferaft distribution, IC stability, naive P&L.
<code>verify.py</code>	The falsification pass. Re-tests Boat Party causally, strips UQ Dollar from the cross-sectional panel, sweeps Fintech windows, and produces the master validated-signal table.
<code>worked_examples.py</code>	Every arithmetic example quoted in this document.
<code>make_figures.py</code>	All ten figures.
<code>build_report.py</code>	Renders this PDF from <code>REPORT.md</code> .

```

pip install numpy pandas scipy statsmodels arch matplotlib markdown weasyprint
cd research
python structure_tests.py && python deep_dive.py && python verify.py
python worked_examples.py && python make_figures.py && python build_report.py

```